PROTOCOL MEMO — PRACTICE 2 **Team 1 · 25 October 2026**

Roles: R1 Diana Luque; R2 Francisco Moya; R3 Claudia Martínez; R4 Fernando Caceres; R5 Marina García.

1. Protocol construction

Cyber uses five-fold StratifiedGroupKFold on feature_group, stratified by attack_code as far as grouping permits. Storage uses five-fold GroupKFold on config_group, keeping all five configuration replicates together. Both grouped splitters and the row-wise comparators shuffle with random_state = 2627. This fixed seed is printed first and was not selected through experimentation.

Cyber contains 34,928 development rows and 34,883 groups; storage contains 12,190 rows and 2,438 configurations. The intended generalisation units are unseen feature groups and unseen storage configurations, respectively.

2. Verified invariants

Every constructed fold has zero shared training/validation groups. Canonical checks confirm that each group belongs to one fold and every row is validated exactly once. All ten cyber classes occur in every validation fold. Ransomware has 208 observations, distributed across canonical folds as 42, 42, 41, 41 and 42. Every storage configuration has five replicates. Any overlap, incomplete coverage or missing validation class constitutes a failed check.

3. Split-protocol comparison

P2.2 compares grouped and row-wise validation using the same fixed five-neighbour KNN and preprocessing. Cyber applies one-hot encoding to three nominal predictors and log1p followed by standardisation to eight numeric predictors. Storage applies one-hot encoding to four nominal predictors and standardisation to five numeric predictors. Learned preprocessing is fitted within each training fold.

Cyber macro F1 is 0.9442 with grouped validation and 0.9443 with row-wise validation: grouped minus row-wise = −0.0001. Storage MAE(z) is 0.2124 versus 0.0809: a difference of +0.1315, with grouped error approximately 2.62 times larger.

Storage's row-wise estimate is optimistic because training can contain zero-distance replicates of validation configurations. Cyber's observed gap is small; grouping remains necessary to measure generalisation to unseen groups. Fold numbers from different protocols do not identify identical validation observations. Fold dispersion is descriptive, not a confidence interval or significance test.

4. Target representation

From P2.3 through P5, storage models use z = log10(latency_ns). Latency spans approximately 4.3 orders of magnitude, from 1.16×10^5 to 2.53×10^9 ns. The raw mean is about 23 times the median; skewness decreases from approximately 9 to 1 after transformation. The logarithm reduces the influence of extreme latencies without assuming a normally distributed target.

MAE(z) measures multiplicative deviations: $|z - \hat{z}| = |\log10(\text{predicted_ns} / \text{observed_ns})|$. The same multiplicative error therefore receives the same cost at different latency levels. Replicate variability also becomes less dependent on latency: its typical standard deviation increases approximately 1,300-fold across configuration quintiles in nanoseconds, compared with ninefold in z.

This choice is not based solely on predictive improvement. On the same canonical folds, five-neighbour KNN achieves MAE(z) ≈ 0.2103 when trained on raw latency and 0.2091 when trained on z. A physical metric is retained because the slowest fifth accounts for approximately 92% of raw-nanosecond error in this diagnostic.

5. Metric contract

Cyber's primary metric is macro F1, supported by weighted F1, accuracy, per-class precision, recall and F1 with support, and a semantically labelled confusion matrix. Macro F1 gives each class equal weight. Ransomware analysis considers missed attacks and false-alarm workload using out-of-fold probabilities, ROC/AUC, Brier score and the five prescribed thresholds. Ten-class argmax is a separate reference.

Storage's primary metric is MAE(z), supported by RMSE(z), R^2(z) and MAE in nanoseconds. Physical errors are calculated by first converting each prediction through predicted_ns = 10^ẑ, then comparing it with the original latency. The quantity 10^MAE(z) is a multiplicative factor, not an error in nanoseconds or a guaranteed coverage bound. Converted predictions are checked for finiteness and strict positivity.

6. Frozen predictive-information basis

The primary route retains all eleven cyber predictors and all nine storage predictors, recorded in plantillas/features.md. Identifiers and grouping metadata—row_id, feature_group, config_group and fold_id—never enter an estimator. Excluded iops is not reintroduced. P2 adds no derived predictive features or extension rows.

Later model families may use justified representations of the same information. Comparisons concern complete pipelines unless representations are held constant or earlier models are rerun. Source information, canonical folds and primary metrics remain fixed. Development scores used for selection are not independent external-test estimates.

7. Canonical adoption and reproducibility

Independent folds and invariant checks precede canonical-file loading. Cyber reproduces the canonical partition exactly; storage differs in configuration membership even after relabelling, while preserving grouping and fold sizes. No alternative seed was searched. All subsequent scored results use the course canonical assignments.

Both final notebooks were executed in fresh kernels with saved outputs and no errors. Both development-only submission dry runs returned "valid"; storage additionally checks physical magnitude and positivity. No final-test features were read. The recorded environment is Python 3.12.14, NumPy 2.5.4, pandas 3.0.1, scikit-learn 1.9.1 and matplotlib 3.11.2.

Data attribution and references

Cyber derives from UNSW TON_IoT Network data; storage derives from HSE-LAMBDA's Data Storage System Performance dataset, Apache-2.0: *Performance Modeling of Data Storage Systems Using Generative Models*, IEEE Access 13 (2025), 49643–49658, doi:10.1109/ACCESS.2025.3552409. Full provenance and terms are recorded in the corresponding PROVENANCE.md files.

References

Full bibliographic details and dataset terms are provided in the corresponding PROVENANCE.md files.

[1] Moustafa, N. Network TON_IoT datasets. *Sustainable Cities and Society*, 2021, 102994.
[2] Booij, T. M., et al. ToN IoT: heterogeneity and standardisation. *IEEE Internet of Things Journal*, 2021.
[3] Alsaedi, A., et al. TON_IoT telemetry dataset. *IEEE Access*, 8, 2020, pp. 165130–165150.
[4] Moustafa, N., et al. Federated TON_IoT Windows datasets. *TrustCom*, 2020, pp. 848–855.
[5] Moustafa, N., Ahmed, M., and Ahmed, S. ToN_IoT Linux datasets. *TrustCom*, 2020, pp. 727–735.
[6] Moustafa, N. TON_IoT datasets for cybersecurity applications. *eResearch Australasia*, 2019.
[7] Moustafa, N. IoT-Fog-Cloud architecture: a review. *arXiv:1906.01055*, 2019.
[8] Ashraf, J., et al. IoTBoT-IDS. *Sustainable Cities and Society*, 2021, 103041.